

L1 — Motion Models and Dead Reckoning

11765 · Navigation and Environment Modelling in Mobile Robotics · MUSI

Lab · 2 October 2026 · Guided lab (not graded)

In this lab, you will explore a real robotics dataset, implement two velocity motion models and evaluate how dead-reckoning errors evolve.

1 Access the course repository

Open the [MUSI-labs GitHub repository](#). Clone it to your computer (or pull the latest changes if you already have it) and locate the L1_dead_reckoning notebook. Read the repository README before starting.

2 Explore the MRCLAM dataset website

Before running the notebook, visit the [official UTIAS MRCLAM dataset website](#). Explore the experiment setup and file descriptions. Identify what the odometry, landmark measurements and ground-truth files contain, and consider which information a real robot can access online versus what is available only for evaluation.

3 Set up your environment and data

Follow the installation and dataset instructions in the repository README. Install the required software and Python dependencies. Launch Marimo from the repository root so the data-loading code can find the files.

4 Complete the L1 notebook

Work through the notebook in order. Start by exploring the data and sampling rates; then implement *wrap_angle*, *motion_euler*, *motion_exact* and *position_errors*. Run the dead-reckoning and subsampling experiments, compare different robots and datasets, and complete every ANSWER section.

What to keep

Save your completed notebook, including your code and written answers. You will reuse the four functions in E1. This lab is formative and it is not graded.